

6(a)	any four from: <i>inheritance is through mtDNA:</i> 1 MELAS syndrome is not inherited from affected males ; 2 all offspring of a female with MELAS syndrome also have MELAS syndrome ; 3 ref. to numbered individuals to support mp1 or mp2 ; <i>if X-linked:</i> 4 males more likely to be affected ; 5 (because) males inherit only one X chromosome (from mother) ; 6 (however) more females affected than males ; 7 there are no, heterozygous / carrier, females ;	4
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Question	Answer	Marks
6(b)	any three from: <i>in mitochondrial DNA:</i> 1 mutations occur at constant rate ; 2 mutations occur at faster rate than, nuclear / chromosomal, DNA ; 3 not protected by histone proteins ; 4 no enzymes to repair DNA mutations ; 5 many copies of mtDNA per cell ; 6 no mixing of DNA at fertilisation (as only inherited from mother) or circular DNA, so no crossing over (all sequence changes are mutations) ;	3

Question	Answer	Marks
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